

26E5610



**DWARAKA DOSS GOVERDHAN DOSS VAISHNAV COLLEGE
(AUTONOMOUS)**

College with Potential for Excellence
Linguistic Minority Institution. Affiliated to University of Madras

**PG DEPARTMENT OF COMPUTER SCIENCE
CIA I – AUGUST 2026
DATA COMMUNICATION AND NETWORKING**

(20) copies

Class: I M.Sc(CS)

Maximum Marks: 50

DATE: 18.08.26

Section-A (10 * 1 = 10 marks)

Answer all the questions

1. What is Computer Network?
2. List any two Topology of computer networks.
3. Write the function of Simplex protocols
4. Define Frame.
5. Give example for BRI and PRI. ✓
6. Define frame.
7. What is meant by Piggybacking? ✓
8. Write any two functions of data link layer?
9. Define checksum. ✓
10. Expand WDM.

Section-B (4 * 5 = 20 marks)

Answer all the questions

11. a) Discuss about Network Hardware by scale.
Or
b) Write a short note on TCP/IP model.
12. a) Differentiate Connection-oriented and Connection Less Service.
Or
b) Write short note on Coaxial and Fiber Optics Cable.
13. a) Explain TDM.
Or
b) Discuss the structure of Telephone System.
14. a) Write short note on CRC.
Or
b) Discuss design issue of datalink layer.

Section-C (2 * 10 = 20 marks)

Answer ALL the questions

15. Explain OSI Reference Model in detail.
16. a) Discuss Unguided Transmission media.
Or
b) Elaborate various Multiplexing Techniques.



26 E 5610

DWARAKA DOSS GOVERDHAN DOSS VAISHNAV COLLEGE
(AUTONOMOUS)
College with Potential for Excellence
Linguistic Minority Institution. Affiliated to University of Madras

PG DEPARTMENT OF COMPUTER SCIENCE
CIA I – AUGUST 2026
ADVANCED SOFTWARE ENGINEERING

I M.Sc.(CS)
SUBCODE: 2627105(A)

DATE: 19.8.2026
Max. Marks: 50

SECTION A (10*1=10 marks)
Answer ALL questions

1. What is the software?
2. Name any two real time software applications
3. What is a waterfalls models?
4. What is estimations?
5. What is LOC & encapsulation?
6. Which software model is more applicable for student database creations?
7. Define quality in software?
8. What is standard?
9. What is a quality matrix?
10. What is the difference between a planning and a scheduling?

SECTION B (4*5=20 marks)

- 11.a) Explain the software life cycle model
OR
b) Determine the RAD and Agile model with example
- 12.a) Demonstrate any two project management models with an example.
OR
b) Explain software quality with an example.
- 13a) Define scheduling & types and Explain with an example.
OR
b) Write a Short note on Polymorphism with an example.
- 14a) Explain Encapsulation, Abstraction, Inheritance with examples
OR
b). Discuss about object modelling with an Example.

SECTION C (2*10=20 marks)

- 15.Explain types of inheritance with an example program.
- 16.(a) Briefly explain the concept of Process matrix & Product matrix with example
- (or)
- (b) Briefly explain the concept COCOMO Model with architecture and example



**DWARAKA DOSS GOVERDHAN DOSS VAISHNAV COLLEGE
(AUTONOMOUS)**

College with Potential for Excellence
Linguistic Minority Institution. Affiliated to University of Madras

**PG DEPARTMENT OF COMPUTER SCIENCE
CIA I – AUGUST 2026
CONTEMPORARY WEB TECHNOLOGIES**

Class: I M.Sc
Subject Code: 2527103

Maximum Marks: 50
Date: 17.08.2026

Section-A (10 * 1 = 10 marks)

Answer all the questions

1. List the basic tags of HTML.
2. Define HTML.
3. Discuss the types of HTML tags.
4. What is HTML5?
5. List the media tags in HTML5.
6. What is JavaScript?
7. List the looping statements available in JavaScript
8. What is the purpose of the document.getElementById() method in JavaScript?
9. What is form validation in JavaScript?
10. What is DOM?

Section-B (4 * 5 = 20 marks)

Answer all the questions

11. a) Differentiate HTML and HTML5.
Or
b) Explain image tag with example.
12. a) Discuss hyperlink in HTML with examples.
Or
b) Explain the features of HTML5
13. a) Describe JavaScript functions .
Or
b) Explain the datatypes and literals of JavaScript with examples.
14. a) List and explain the advantages of JavaScript.
Or
b) Differentiate between Built-in Objects and User-Defined Objects in JavaScript. .

Section-C (2 * 10 = 20 marks)

Answer ALL the questions

15. Explain in detail about the new input elements in HTML5 with its attributes and example.
16. a) Explain different event handling methods in JavaScript with examples.
Or
b) Elaborate List in HTML with example program.



26E5610

DWARAKA DOSS GOVERDHAN DOSS VAISHNAV COLLEGE
(AUTONOMOUS)
College with Potential for Excellence
Linguistic Minority Institution. Affiliated to University of Madras

PG DEPARTMENT OF COMPUTER SCIENCE
C.I.A I - August 2026
Python for Data Science

Class: I M.Sc.
Subject Code : 2627102
Date: 14.08.2026

Maximum Marks : 50 Marks
Duration: 2 Hrs

Section-A (10 X 1 = 10 Marks)

1. Define the variable and keyword.
2. How to write single and multiline comments in python?
3. List the rules for naming the identifier.
4. Give an example for type casting.
5. State the significance of indentation.
6. What is meant by type ()?
7. Define the expression.
8. How to do multiple assignments in python?
9. Write the Operator precedence and how to override it.
10. List the comparison operators.

Section-B (4 X 5 = 20 Marks)

- 11) a) List out the features of Python and justify why Python is referred to as dynamically typed?
or
b) Give an overview of the Datatypes.
- 12) a) List the built in functions used for String manipulation.
or
b) Brief a note on input and output statements with example.
- 13) a. Differentiate the break, pass and continue statements.
or
b) Explain how you use different modules in python.
- 14) a) Discuss in detail about Lambda functions.
Or
b) Explain the concept of Arrays in Python.

Section-C (2 X 10 = 20 Marks)

- 15) Write in detail about Lists, Dictionaries, Sets and Sequences.
- 16) a) Describe the different types of functions.
or
b). Elaborate Python conditional and looping statements.



26E5610

DWARAKA DOSS GOVERDHAN DOSS VAISHNAV COLLEGE
(AUTONOMOUS)
College with Potential for Excellence
Linguistic Minority Institution. Affiliated to University of Madras

PG DEPARTMENT OF COMPUTER SCIENCE
CIA I – AUGUST 2026
ADVANCED DESIGN AND ANALYSIS OF ALGORITHMS

Class: I M.Sc. (Computer Science)
Subject Code: 2527101

Max. Marks: 50
Date: 13.08.2026

SECTION A (10*1=10 marks)
Answer ALL questions

1. What is an Algorithm?
2. What are the metrics to evaluate the performance of an algorithm?
3. Write any two characteristics that an algorithm must satisfy.
4. Define the Binary tree.
5. What are the two representations of graph data structure?
6. What is the divide and conquer method?
7. Define min heap and max heap
8. Write about the selection sort method
9. Write algorithm for Heap sort
10. How the operations performed in Strassen's Matrix multiplication

SECTION B (4*5=20 marks)

11. a) Tower of Hanoi problem is solved using a recursive algorithm. Discuss
(or)
b) Explain in brief about operations performed in Trees.
12. a) Discuss operations performed in Queues explain with algorithm.
(or)
b) Explain in brief about Asymptotic Notations.
13. a) Explain Kruskal's Minimum cost spanning tree algorithm with a suitable example.
(or)
b) Explain the selection sort in detail. Illustrate the algorithm with a numeric example.
14. a) Explain the Radix sort in detail. Illustrate the algorithm with a numeric example..
(or)
b). Explain the merge sort in detail. Illustrate the algorithm with a numeric example.

SECTION C (2*10=20 marks)

15. Explain Prim's algorithm for the minimal spanning tree with an example. (K4)
16. a) Discuss in detail the working of stack data structure. Analyze the operations performed in the stack with an algorithm. (K4)
(or)
b) Explain the quicksort algorithm to sort 'n' numbers with suitable example.